Duration: 90 min

...Fall 25/26

Q1: choose the correct answers:

(Please write your answers in the table only; otherwise, your answers will be neglected) **(15 marks)**

Question	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*
Answer										

1. The twisted pair cable in which metal casing improves the penetration of noise or crosstalk is called
   - A. insulated twisted pair cable
   - B. Shielded twisted pair cable
   - C. Unshielded twisted pair cable
   - D. Both A and B

2. Maltiplexing combincs signals from several sources to achieve
   - A. Data rate management
   - B. Interleaving
   - C. TDM efficiency
   - D. bandwidth efficiency

3. The synchronous Time-Division Multiplexing (TDM) is not efficient because of
   - A. higher data rate
   - B. infinite frames
   - C. empty slots
   - D. Both A and C

4. The multiplexing technique that is used when the data rate of an input line is a mnltiple of the others is called
   - A. multilevel mnltiplexing
   - B. multiple-slot allocation
   - C. pulse stuffing
   - D. Asynchronous TDM

5. TDM is a digital multiplexing technique for combining several low-rate channels into
   A. a group of low-rate channels
   B. one high-rate one channel
   C. two high-rate channels
   D. None

6. CRC stands for
   A. combine resistance check
   B. cyclic redundancy code
   C. combine redundancy code
   D. cyclic redundancy check

7. There is no error in the received code word, if the syndrome is
   A. maximum
   B. minimum
   C. 0
   D. 1

8. To guarantee the detection of up to s errors in all cases, the minimum hamming distance in a block code must be
   A. *s*
   B. *s*+1
   C. *s*-1
   D. 0

9. Radio waves are
   A. unidirectional
   B. omnidirectional
   C. bidirectional
   D. directional

10. Optical fibers use reflection to guide light through a
   A. channel
   B. metal wire
   C. light
   D. plastic

Q2: Ten sources, six with a bit rate of 200 kbps and four with a bit rate of 400 kbps, are to be combined using multilevel TDM with no synchronizing bits. Answer the following questions about the final stage of the multiplexing:

a. What is the size of a frame in bits?
b. What is the frame rate?
c. What is the duration of a frame?
d. What is the data rate

(12 marks)

Q3: Draw the constellation diagram for the following cases. The numbers in parentheses define the values of I and Q, respectirely.

a. Two points at (2, 0) and (3, 0)
b. Two points at (3, 0) and (−3, 0)
c. Four points at (2, 2), (−2, 2), (−2, −2), and (2, −2)
d. Two points at (0, 2) and (0, −2)

(12 marks)